

Q1. What is the difference between a Basic Filter, Advanced Filter, and a Slicer in Power BI?

Basic Filter

A **Basic Filter** is the simplest way to filter data.

- Works like a checklist of values
- You manually select/deselect items
- Best for quick filtering of categorical data

Example:

Filtering a “Product Category” field to show only *Electronics* and *Furniture*.

Easy to use

Not suitable for complex conditions

Advanced Filter

An **Advanced Filter** allows more complex filtering logic.

- Use conditions like:
 - *is greater than*
 - *contains*
 - *starts with*
- Can combine multiple conditions (AND / OR)

Example:

Show products where:

- Sales > 10,000 **AND**
- Product name contains “Pro”

Powerful and flexible

Slightly more complex to set up

Slicer

A **Slicer** is a *visual filter* placed directly on the report canvas.

- Interactive UI element (buttons, dropdowns, sliders)
- End users can click to filter visuals in real time
- Great for dashboards and reports

Example:

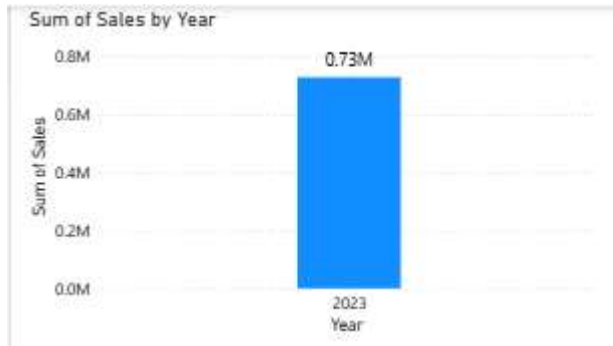
A slicer for *Year* where users click 2023 or 2024 to update charts.

User-friendly and interactive

Enhances report usability

Takes up report space

Q2. Using the Sales dataset, apply a Basic Filter to display only the sales data for the year 2023?



Q3. Create a Pie Chart of Sales by Product and add Slicer that allows users to filter the dataset by region?



Q4. What is a Visual-Level Filter, and how does it differ from Page-Level and Report-Level Filters?

Visual-Level Filter

A **Visual-Level Filter** applies **only to a single visual** (chart, table, card, etc.).

- Located in: *Filters on this visual*
- Affects: **Just one visual**
- Other visuals remain unchanged

Example:

You show a bar chart of sales but filter it to display only **2023**, while other visuals still show all years.

Very precise control

Useful for highlighting specific insights.

Page-Level Filter

A **Page-Level Filter** applies to **all visuals on a single report page**.

- Located in: *Filters on this page*
- Affects: **Every visual on that page**
- Does not impact other pages

Example:

Filter the entire page to show only **Region = West**.

Keeps one page focused on a specific segment

Easier than applying the same filter to multiple visuals

Report-Level Filter

A **Report-Level Filter** applies to **every page and every visual in the report**.

- Located in: *Filters on all pages*
- Affects: **Entire report**
- Global filtering logic

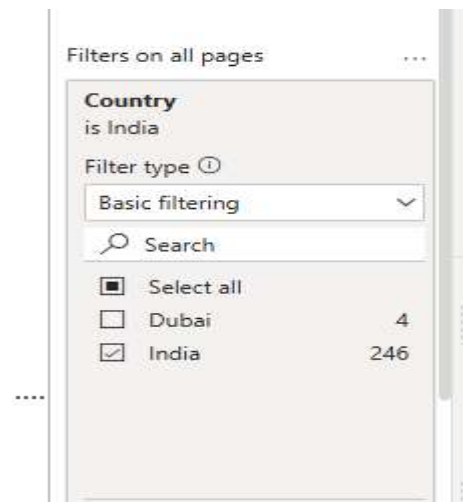
Example:

Set **Year = 2023** for the whole report.

Ensures consistency across all pages

Less flexibility for individual pages

Q5. Apply a Report-Level Filter so that all visuals in the Power BI report display sales only from India?



Q6. What is the role of Drill-Down in Power BI? How does it improve data exploration?

In Microsoft Power BI, Drill-Down is a feature that lets you navigate through different levels of data hierarchy within a visual, moving from a summary view to more detailed levels.

Drill-Down allows you to:

- Start with high-level data (e.g., Year)
- Click into progressively deeper levels (e.g., Quarter → Month → Day)

It works when your data has a hierarchy, such as:

- Date: Year → Quarter → Month → Day
- Geography: Country → State → City
- Product: Category → Subcategory → Product

Example : Imagine a sales chart:

- Initially shows Total Sales by Year
- You click on 2023
- It drills down to Quarter-wise sales
- Click again → Monthly sales
- And so on

Role of Drill-Down

- Enables step-by-step exploration of data
- Helps identify patterns and trends at different levels
- Reduces the need for multiple separate visuals
- Keeps reports clean and interactive

How It Improves Data Exploration

1. From Summary to Detail

Instead of overwhelming users with all data:

- Start high-level
- Explore deeper only when needed

Example: Spot a sales drop in 2023 → drill down to find which month caused it

2. Faster Root Cause Analysis

Quickly identify why something happened

Example:

Low sales → Drill to region → Drill to city → Find underperforming locations

3. Better User Experience

- Interactive and intuitive
- Users control what they want to explore
- No need to switch between multiple reports

4. Saves Space in Reports

Instead of creating:

- 4 visuals (Year, Quarter, Month, Day)
- You use one visual with drill-down

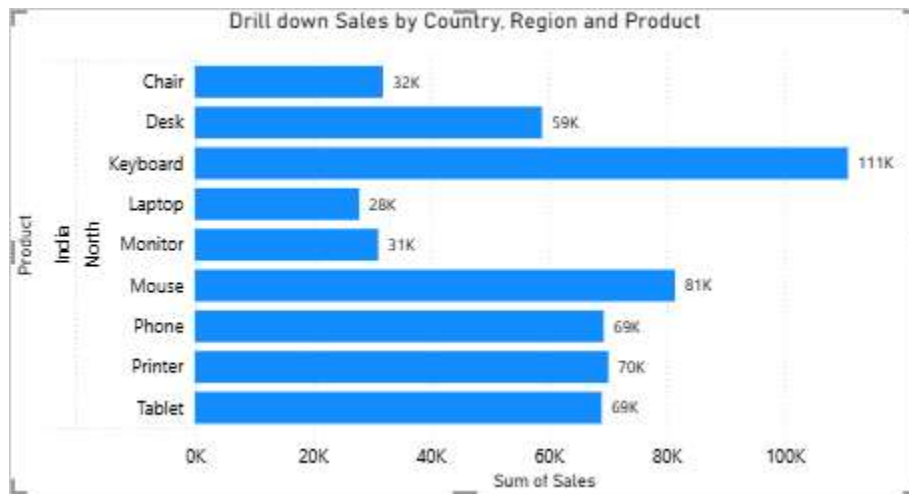
Drill-Down vs Drill-Up

- Drill-Down → Go deeper into details
- Drill-Up → Go back to summary level

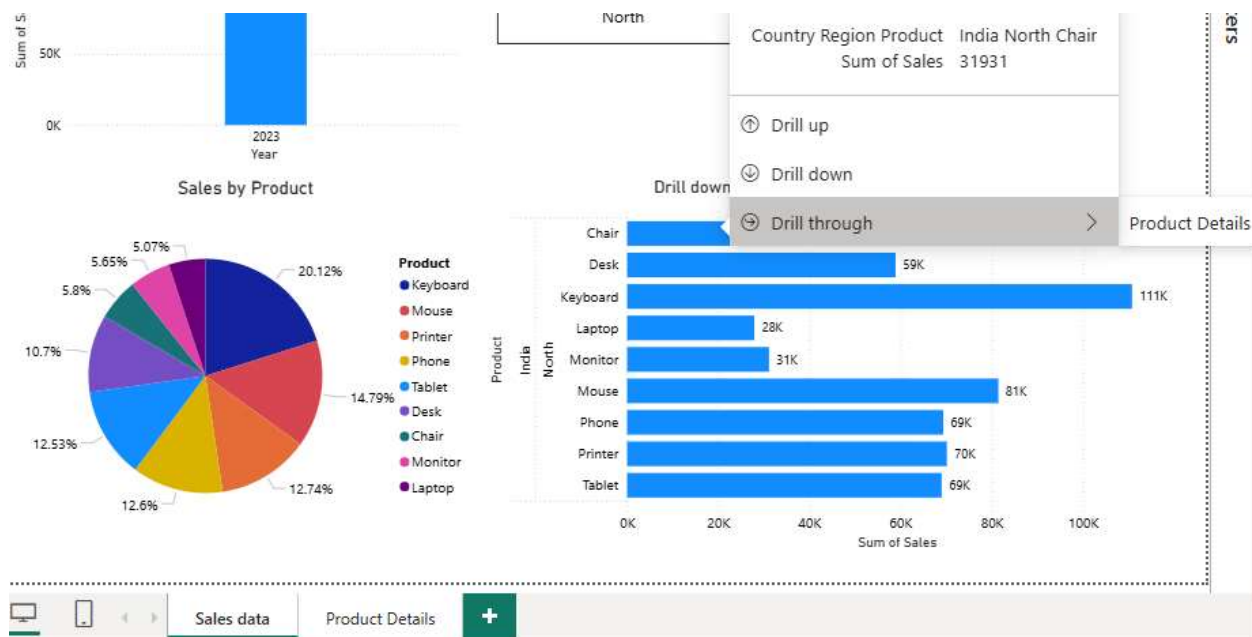
Quick Summary

- Drill-Down = Explore deeper levels of data hierarchy
- Helps users analyze, investigate, and understand data faster

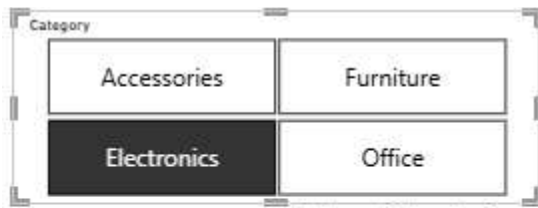
Q7. Using the dataset, create a hierarchy drill-down from Country → Region → Product Category to analyze sales at different levels?



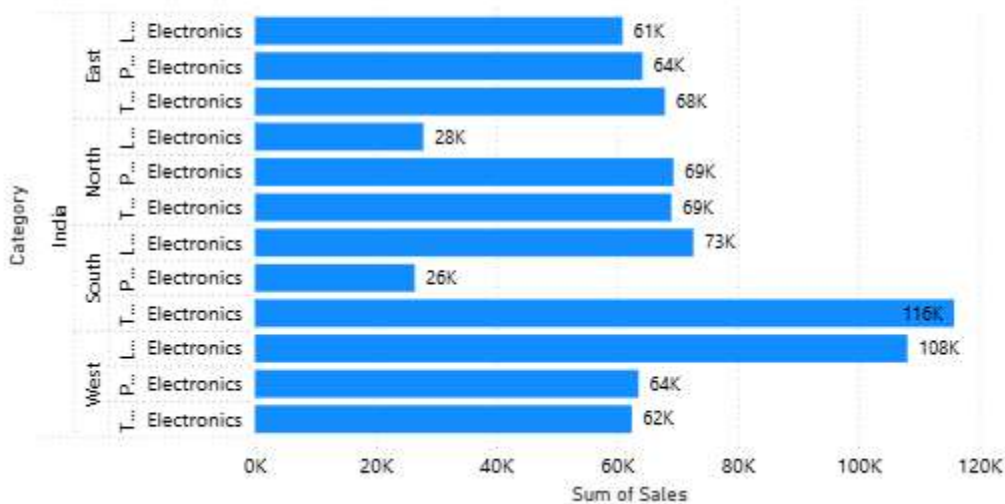
Q8. Implement Drill-Through so that selecting a Product navigates to a detailed report page showing region wise sales?



Q9. Add a Slicer for Product Category and observe how it interacts with the drill-down hierarchy?



Drill down Sales by Country, Region and Product



Q10. Explain the difference between Drill-Down and Drill-Through in Power BI using a real-world example?

Drill-Down Definition: Drill-Down lets you navigate through levels of a hierarchy within the same visual. You move from a summary view to more detailed data without leaving the page.

Real-World Example:

Imagine you have a sales dashboard:

- Visual: Total Sales by Country
- Drill-Down hierarchy: Country → Region → Product Category

Scenario:

- You see that India has high sales.
- Click India → visual drills down to show sales by region in India.
- Click a region → visual drills down further to show sales by product category in that region.

Key point: You stay on the same page, exploring levels of the same data hierarchy.

Drill-Through Definition: Drill-Through lets you jump to a different report page that shows detailed data based on the selected item.

Real-World Example:

Continuing with the same dashboard:

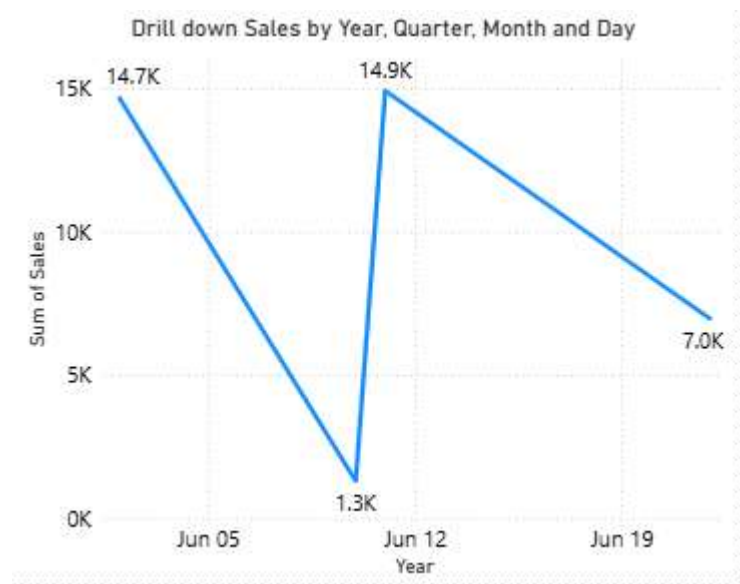
- Main page: Product-wise sales summary
- Drill-Through page: Region-wise sales for the selected product

Scenario:

- On the main page, you right-click Product A → select Drill Through → Product Details page
- The new page shows all regions and their sales for Product A
- You can now analyze where Product A performs best

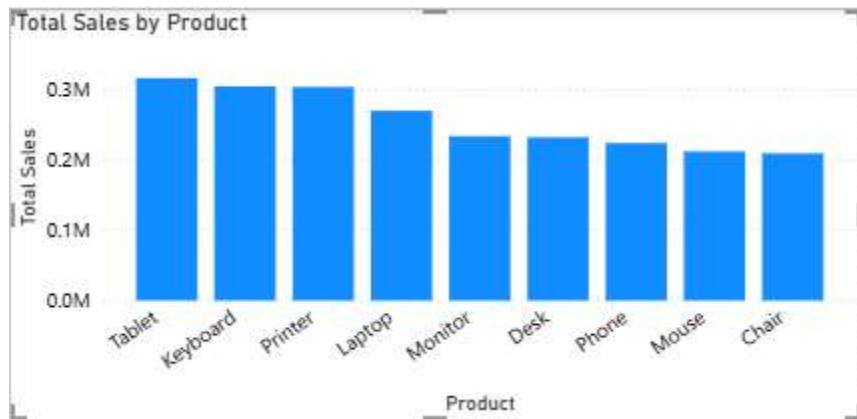
Key point: You navigate to a separate page with detailed visuals based on the selected context.

Q11. Use the Drill-Down feature to analyze sales trends from Year → Quarter → Month → Day?



Q12. Apply a Measure-Based Filter to display only those products where total sales exceed ₹50,000. Hint: Create a Total Sales measure and apply the filter using the measure, not the Sales column.

```
1 Total Sales = sum(Sheet1[Sales])
```



Q13. What is the “Keep All Filters” option in Drill-Through? How does it affect user interaction?

When you create a Drill-Through page, you specify one or more Drill-Through fields (e.g., Product, Year).

- By default, only the Drill-Through field filters are passed to the target page.
- “Keep All Filters” is an option you can enable on the Drill-Through page.

Purpose:

It ensures that all other filters applied on the source page (like slicers, page-level filters, or visual-level filters) are also passed to the Drill-Through page.

How It Works

Without “Keep All Filters”

- right-click Product A → Drill Through
- Only Product A context is passed to the target page
- Other slicer or filter selections (like Region = West, Year = 2023) are ignored

With “Keep All Filters” enabled

- Right-click Product A → Drill Through
- Product A context is passed plus all other active filters
- Target page respects selections like:
 - Region = West
 - Year = 2023
 - Customer Segment = Retail

Effect on User Interaction

1. Preserves Context: Users don't lose other filter selections when drilling through.
2. Consistent Experience: Keeps drill-through page aligned with the source page view.
3. Flexible Analysis: Users can slice data on multiple dimensions and drill-through without losing those filters.
4. Potential Pitfall: If too many filters are active, the drill-through page may show very few or no data.

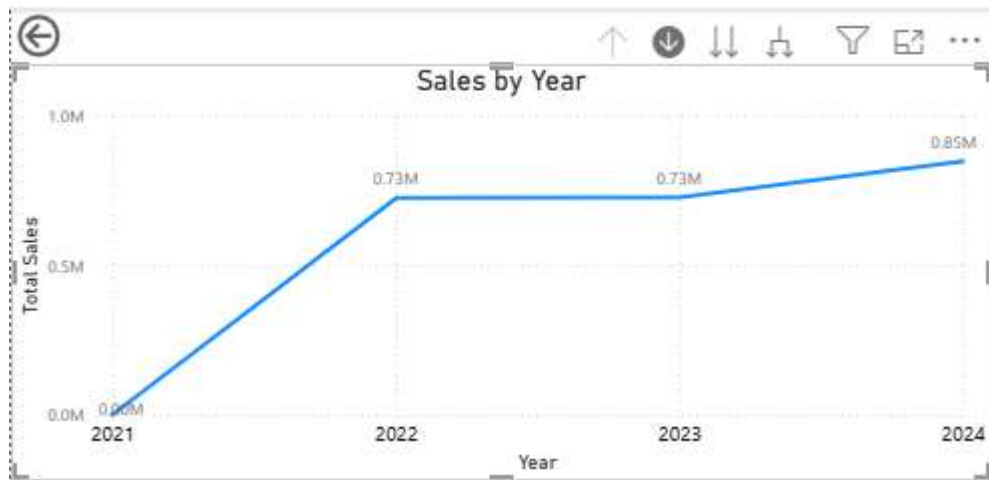
Real-World Example

- Main Page: Product-wise sales with slicers for Year and Region
- Drill-Through Page: Region-wise sales for the selected product

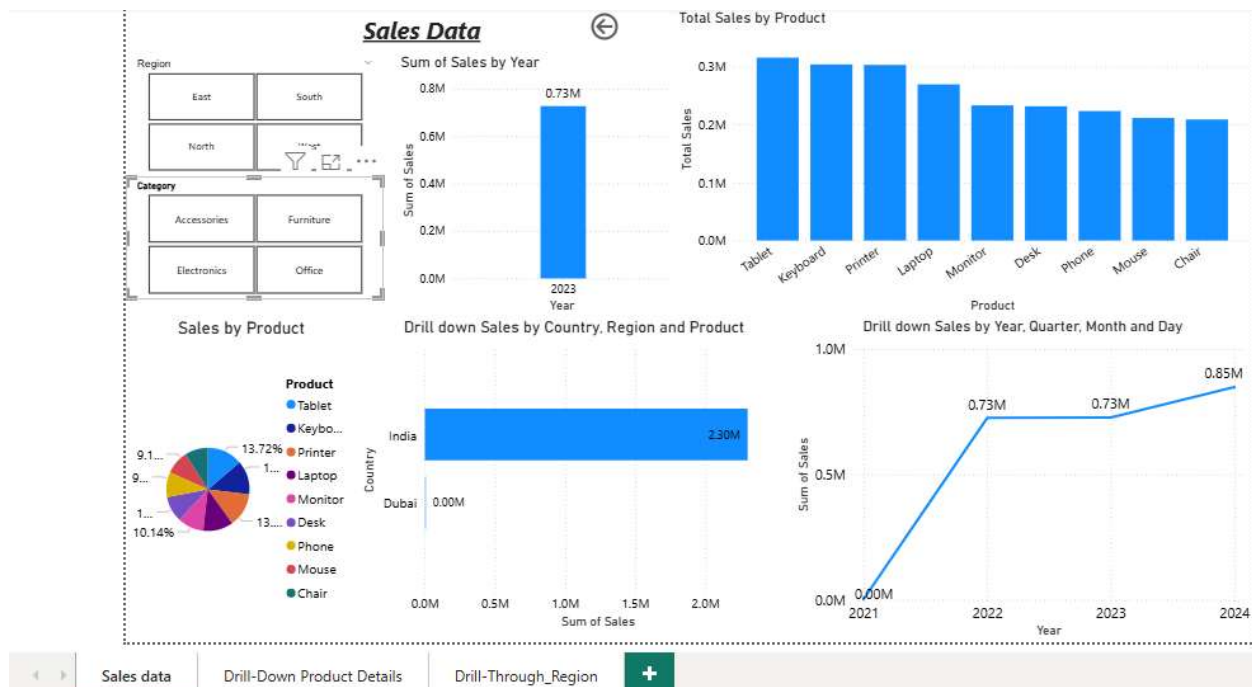
Scenario:

- User selects: Year = 2023, Region = West
- Right-click Product A → Drill Through → Region-wise sales page
- With Keep All Filters ON:
 - Page shows Product A sales in 2023, West region only
- With Keep All Filters OFF:
 - Page shows Product A sales across all regions and years

Q14. Create a custom Drill-Through page where selecting a Region shows a detailed view of sales trends overtime?

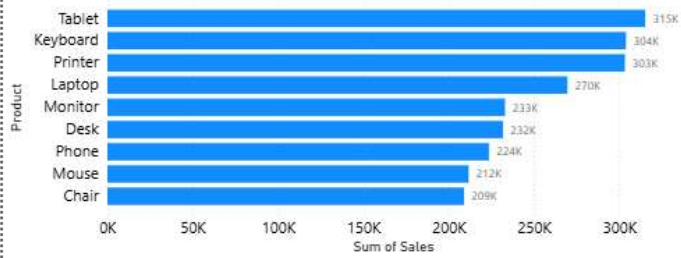


Q15. Design a fully interactive Power BI dashboard using Slicers, Filters, Drill-Down, and Drill-Through to explore sales data?

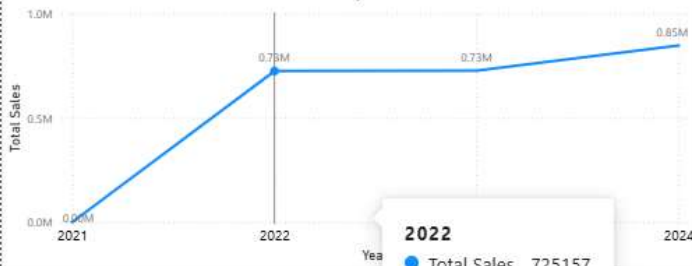




Sum of Sales by Product



Sales by Year



2022

● Total Sales 725157

⌵ Drill down